

Weld Schedule Manager (Root Pass)



WPS No	July test weld 6	Position / Direction	Fixed 5G - Down	Diameter (in)	48	Wall (mm)	13.7
Process	LSC (Modified Short Arc)	Specification	API1104	Filler Wire	Surearc ER70S-6(1.1mm)	Shield Gas	Ar/CO2 85/15
Bevel Design	30 degree	Material Grade	X70	Trade Name	Sure-arc	Completed by	Ben Hatch

0°	Travel Speed	Torch Angle	Dynamic Correction
0mm	14	-20	-2
1mm	12	10	-1
2mm	7	10	0
3mm	4.5	10	1
4mm	3	10	2
5mm	2	5	3

30°	Travel Speed	Torch Angle	Dynamic Correction
0mm	15	-15	-2
1mm	13	0	-1
2mm	9	0	0
3mm	6.5	0	0
4mm	4.5	-5	0
5mm	2.75	-5	0

60°	Travel Speed	Torch Angle	Dynamic Correction
0mm	15.5	-10	-2
1mm	13.5	-5	-1
2mm	10	-10	0
3mm	7.5	-15	0
4mm	5	-15	0
5mm	2.75	-15	0

90°	Travel Speed	Torch Angle	Dynamic Correction
0mm	16	-5	-2
1mm	14	-5	-2
2mm	10.5	-10	-2
3mm	8	-15	-1
4mm	5	-15	0
5mm	2.75	-15	0

120°	Travel Speed	Torch Angle	Dynamic Correction
0mm	15.5	0	-3
1mm	13.5	-5	-3
2mm	10	-10	-2
3mm	7.5	-15	-1
4mm	5	-15	0
5mm	2.75	-15	0

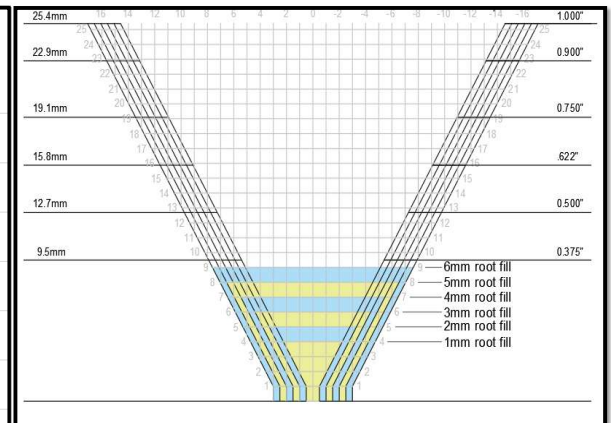
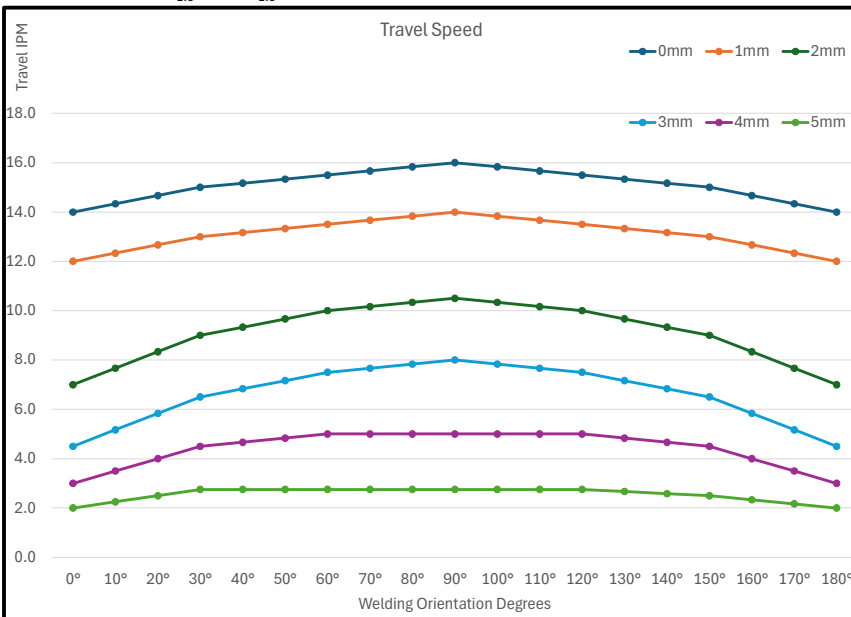
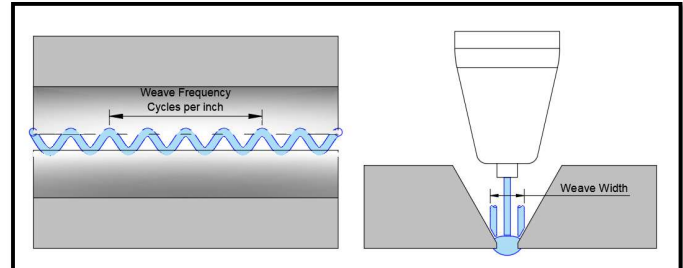
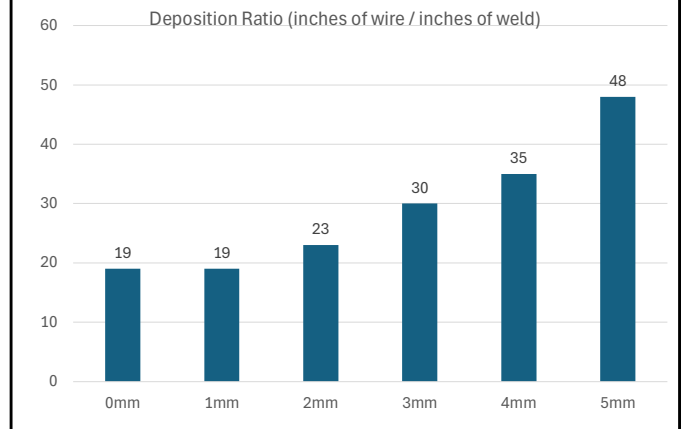
	Cycle Times (mins)	Wire Consumption LBS
0mm	2	0.3
1mm	3	0.3
2mm	4	0.4
3mm	6	0.5
4mm	9	0.6
5mm	15	0.9

Wire Diameter	1.1	
Type of line up clamp	Internal / External	
Min % root before clamp removal	75%	
Max time between bead & HP	24 Hours	
Shield Gas Flow (CFH)	55	
Min Pre-Heat Temp C° / Method	250°	Propane

180°	Travel Speed	Torch Angle	Dynamic Correction
0mm	14	0	-5
1mm	12	0	-5
2mm	7	0	-4
3mm	4.5	0	-3
4mm	3	0	0
5mm	2	0	0

150°	Travel Speed	Torch Angle	Dynamic Correction
0mm	15	0	-4
1mm	13	0	-4
2mm	9	-5	-3
3mm	6.5	-15	-2
4mm	4.5	-15	0
5mm	2.5	-15	0

Deposition Adjustment			CTWD	Arc Length	Weave Type	Weave Frequency	Weave Width	Weave Length	Weave Dwell
Gap	Whole Pipe	Lower Half							
0mm	0%	0%	-3	0	None	0	1.2	1.2	0
1mm	0%	0%	-3	0	None	0	1.2	1.2	0
2mm	0%	0%	-2	0	Circle	12	1.4	1.4	0
3mm	0%	0%	0	0	Circle	12	3.6	3.6	0
4mm	0%	0%	1	0	Circle	12	5.2	3.8	0
5mm	20%	10%	2	0	Circle	12	6.4	4.8	0



Cleaning / Grinding				
Root Gap	Grind Weld Pass	Travel Speed	Weave Amplitude	Frequency Hz
0mm	ON	100	3	2
1mm	ON	90	3	2
2mm	ON	90	3	2
3mm	ON	90	4	2
4mm	ON	90	4.5	2
5mm	ON	90	5	2

Notes

Changes: We made the weave parameters the same as they were before June when we changed them trying to use the pipeliner wire. We had to do this because we came close to burning through and eventually did using the surearc. We made the weaves marginally wider on the schedules 3mm and above. (Burned through 3 times on 3 different welds between 150 and 175 degrees). the above parameters includes the changes made.

Results: after the burn through and the changes made to the weave width it welded fine. it is inconclusive if it welded fine as a result of the change or if it was a coincidence. but the changes will remain until we observe a lack of penetration on gaps larger than 3mm.

Weld Schedule Manager (Fill & Cap)

WPS No	July test weld 6	Position	Fixed 5G - Down	Diameter (in)	48	Wall (mm)	13.7
Process	GMAW	Specification	API1104	Filler Wire	Lincoln Pipeliner ER70-S6 (1.2mm)	Shield Gas	Ar/CO2 85/15
Bevel Design	30 degree	Material Grade	X70	Trade Name	Sure-arc	Completed by	Ben Hatch



0° Position	
Travel Speed	15
Torch Angle	0

30° Position	
Travel Speed	16
Torch Angle	-5

60° Position	
Travel Speed	18
Torch Angle	-10

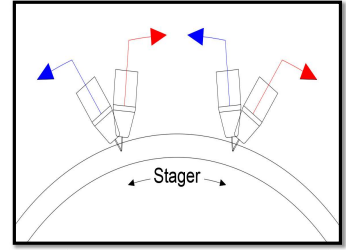
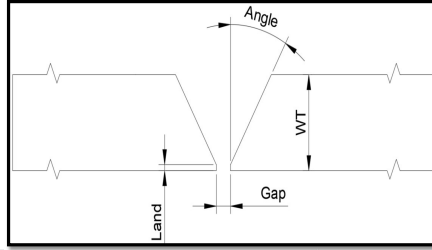
90° Position	
Travel Speed	20
Torch Angle	-10

120° Position	
Travel Speed	18
Torch Angle	-10

150° Position	
Travel Speed	14
Torch Angle	-10

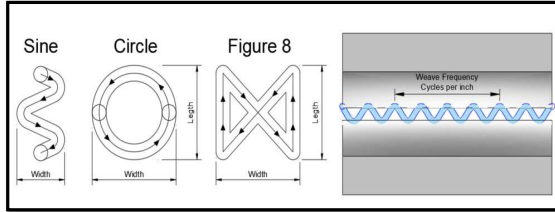
180° Position	
Travel Speed	10
Torch Angle	-10

Wire Diameter	1.2
Direction of Travel	Vertical Down
Process	Pulse
Deposition Ratio	12
Shield Gas Flow (L/min)	26



Joint Geometry		Tolerance (+ / -)	
Wall Thickness (mm)	13.7	1	0.5
Land (mm)	1	0.5	0.5
Root Gap (mm)	3	1	1
Bevel Angle (°)	30	5	5
Hi / Low (mm)		1.5	1.5

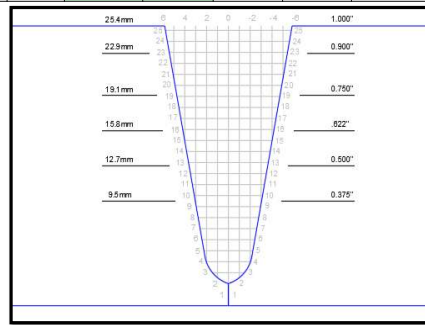
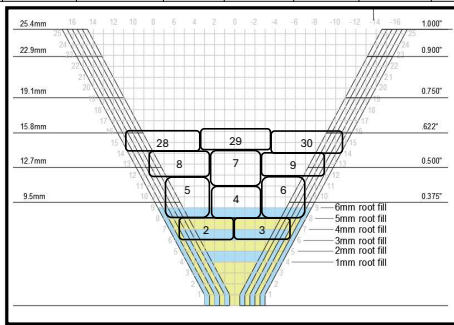
Stop / Start	
Stager the Stop-Start (mm)	30
Over run the next pass (mm)	12
Lead in at re-start (mm)	10



Weld Characteristics	
Deposition Efficacy	0.98
Pass Volume (mm2)	12.5
Min Cap Width (mm)	22
Root Depth (mm)	6
Total Passes (Inc root)	9

Total Arc Time (mins)	21.4
Wire Consumption (lb)	6
Wire Consumption (kg)	3

	Weave Adjustment					Weld Adjustment					Cleaning / Grinding					Comments
	Weave Pattern	Width (mm)	Legth (mm)	Dwell (sec)	Frequency / inch	Deposition Adjustment	Travel Speed Adjustment	Pass Volume	Pass Time	Arc Legth	Grind Prior to Weld Pass	Grind After Weld Pass	Travel Speed (in/min)	Weave Amplitude	Frequency Hz	
Hot Pass 0mm	Sine	1.5	2.0	0.0	12	-20%	-20%	10.6	2.9	2	OFF	ON	75	1	2	
Hot Pass 1mm	Sine	1.5	2.0	0.0	12	-10%	-10%	12.0	2.6	2	ON	OFF	75	1	2	
Hot Pass 2mm	Sine	1.5	2.0	0.0	12	-10%	-10%	12.0	2.6	2	ON	OFF	75	1	2	
Hot Pass 3mm	Sine	1.5	2.0	0.0	12	-10%	-10%	12.0	2.6	2	ON	OFF	75	1	2	
Hot Pass 4mm	Sine	1.5	2.0	0.0	12	0%	0%	13.3	2.3	2	ON	OFF	75	1	2	
Hot Pass 5mm	Sine	2.0	2.0	0.0	12	0%	0%	13.3	2.3	2	ON	OFF	75	1	2	
Fill Passes	Sine	2.0	2.0	0.0	12	0%	0%	13.3	2.3	3	ON	OFF	75	1	2	
Cap Passes	Sine	3.5	3.5	0.0	12	0%	0%	13.3	2.3	4	ON	ON	75	1	2	



Pass #	On/Off	Pass Type	0mm		1mm		2mm		3mm		4mm		5mm		Comments
2	ON	Hot Pass	4.0	2.0	5.0	2.5	5.0	3.0	5.0	3.0	5.0	3.5	6.0	4.0	
3	ON	Hot Pass	4.0	-2.0	5.0	-2.5	5.0	-3.0	5.0	-3.0	5.0	-3.5	6.0	-4.0	
4	ON	Fill Pass	6.0	0.0	8.0	0.0	8.0	0.0	8.0	0.0	8.0	0.0	9.0	0.0	
5	ON	Fill Pass	6.0	3.5	8.0	4.0	8.0	4.5	8.0	4.5	8.0	5.0	9.0	5.5	
6	ON	Fill Pass	6.0	-3.5	8.0	-4.0	8.0	-4.5	8.0	-4.5	8.0	-5.0	9.0	-5.5	
7	ON	Fill Pass	8.0	0.0	9.0	0.0	9.0	0.0	9.0	0.0	9.0	0.0	10.0	0.0	
8	ON	Fill Pass	10.0	4.5	11.0	5.0	11.0	5.5	11.0	5.5	11.0	6.0	12.0	6.5	
9	ON	Fill Pass	12.0	-4.5	13.0	-5.0	13.0	-5.5	13.0	-5.5	13.0	-6.0	14.0	-6.5	
10	OFF	Fill Pass	12.0	0.0	13.0	0.0	13.0	0.0	13.0	0.0	13.0	0.0	14.0	0.0	
11	OFF														
12	OFF														
13	OFF														
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24	OFF														
25	OFF														
26	OFF														
27.0	OFF														
28.0	ON	Cap Pass	13.0	0.0	13.0	0.0	13.0	0.0	13.0	0.0	13.0	0.0	13.0	0.0	
29.0	ON	Cap Pass	13.0	7.0	13.0	7.5	13.0	8.0	13.0	8.0	13.0	8.5	13.0	9.0	
30.0	ON	Cap Pass	13.0	-7.0	13.0	-7.5	13.0	-8.0	13.0	-8.0	13.0	-8.5	13.0	-9.0	

Notes

Changes: We made passes 8 & 9 wider by 0.5mm each side. We made cap passes 29 and 30 wider by 1mm each side. We made the cap sequence go middle then left then right. All to try and balance the fill and make the cap catch both edges comfortably

Results: New Baseline, all changes were effective, the fills were level and the cap was uniform and flat. It is not as aesthetically pleasing to do the center cap pass before the outside passes but it makes the cap flat and symmetrical

Suggestions: repeat test with a push angle on the bottom to see what the difference looks like